

Smart Digital Complaint Management and Public Transparency System

Project Exhibition 1. Review 1.

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Account-less civic issue reporting, public audit trails, and AI-assisted photo validation

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Citizen Reporting Portal

Web portal for citizens to report public infrastructure issues across 10 structured categories — potholes, garbage, water leakage, streetlights, and more.

No Account Required

Email OTP verification only. Citizens submit complaints without registration, reducing friction and maximizing accessibility.

Three-Layer System

Public registry for transparency, admin console for workflow management, and a proposed AI validation layer for automated photo verification.

Problem Identification

Problem Statement: Civic issue reporting is fragmented, untracked, and invisible to the public — no single transparent system captures what was reported, how urgent it is, and what happened next.

Current Reporting Channels

- Phone hotlines
- In-person municipal visits
- Informal social media posts

Systemic Failures

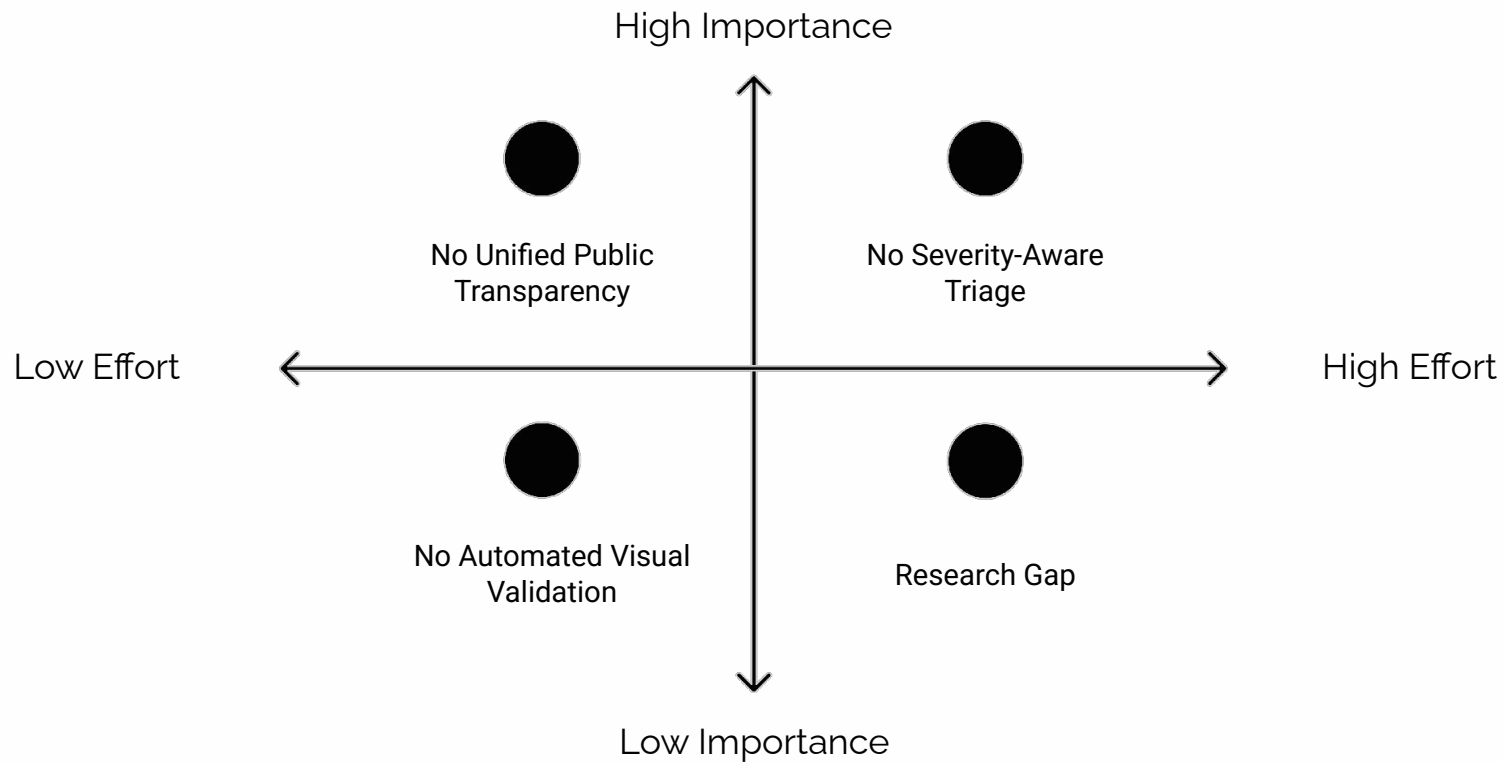
- No tracking ID issued to the citizen
- No public record of complaints filed
- No category-specific urgency signal
- No photo verification mechanism

The absence of any single failure mode would be manageable; the convergence of all four produces a systemic breakdown in civic accountability and public trust.

Literature Survey & Research Gap

A review of prior approaches reveals that no existing solution simultaneously addresses public transparency, severity-aware triage, and automated visual validation.

Prior Approach	Strength	Limitation
Phone hotlines / 311-style systems	Centralized entry point	No photo evidence, no public tracking
Crowdsourced issue maps (e.g. FixMyStreet-style)	Public visibility	No identity check, no urgency scoring
Generic helpdesk ticketing	Mature status workflow	Not citizen-facing / not transparent
CV-based defect detection (research area)	Automated visual check	Needs category-specific training data



Objectives

Accessibility

OTP email verification only — no account creation required, minimizing barriers to civic participation.

Structured Categorization

Fixed 10-category taxonomy ensures consistent classification and enables category-specific handling.

Intelligent Prioritization

Weighted questionnaire score produces an explainable, consistent priority signal for each complaint.

AI Validation (Proposed)

Per-category AI detection model to automate photo verification — proposed architecture, not yet implemented.

Transparency & Auditability

Public registry with full status history ensures every complaint is visible and traceable.

Security

JWT-protected admin actions prevent unauthorized status changes or data manipulation.

Scope

In Scope — Current MVP

- OTP-verified complaint submission
- 10-category structured questionnaire
- Weighted priority scoring
- Public complaint registry
- Admin status workflow
- PDF receipt generation

Out of Scope

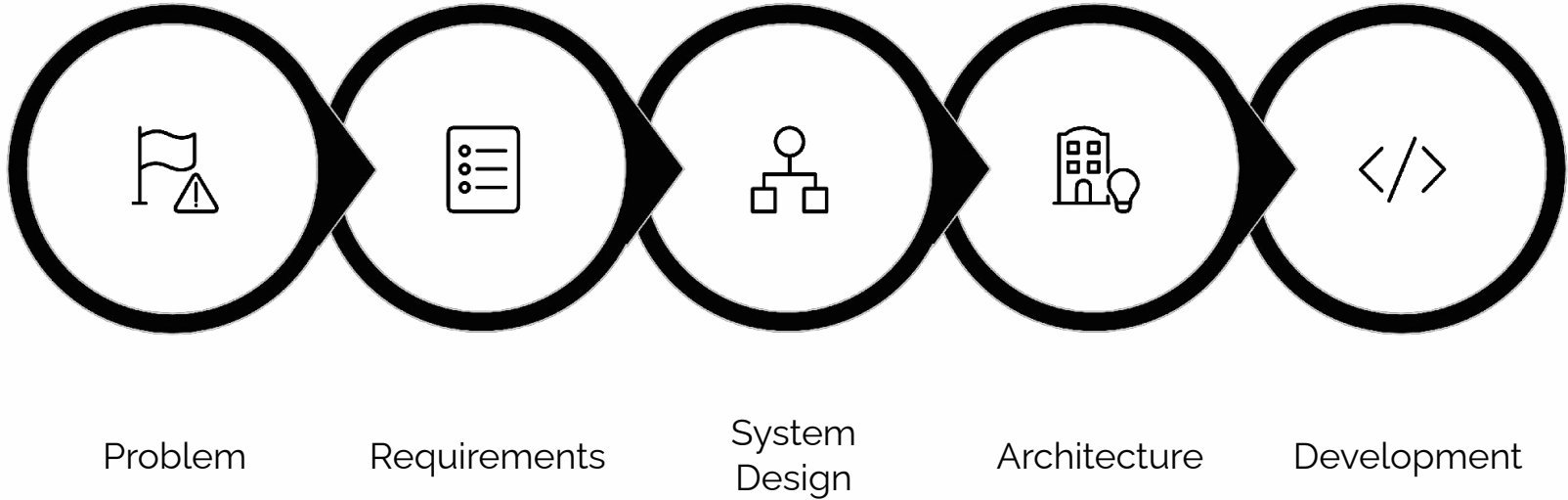
- Native mobile applications
- In-app chat or messaging
- Payment or fee processing
- Multi-language support

Future Scope

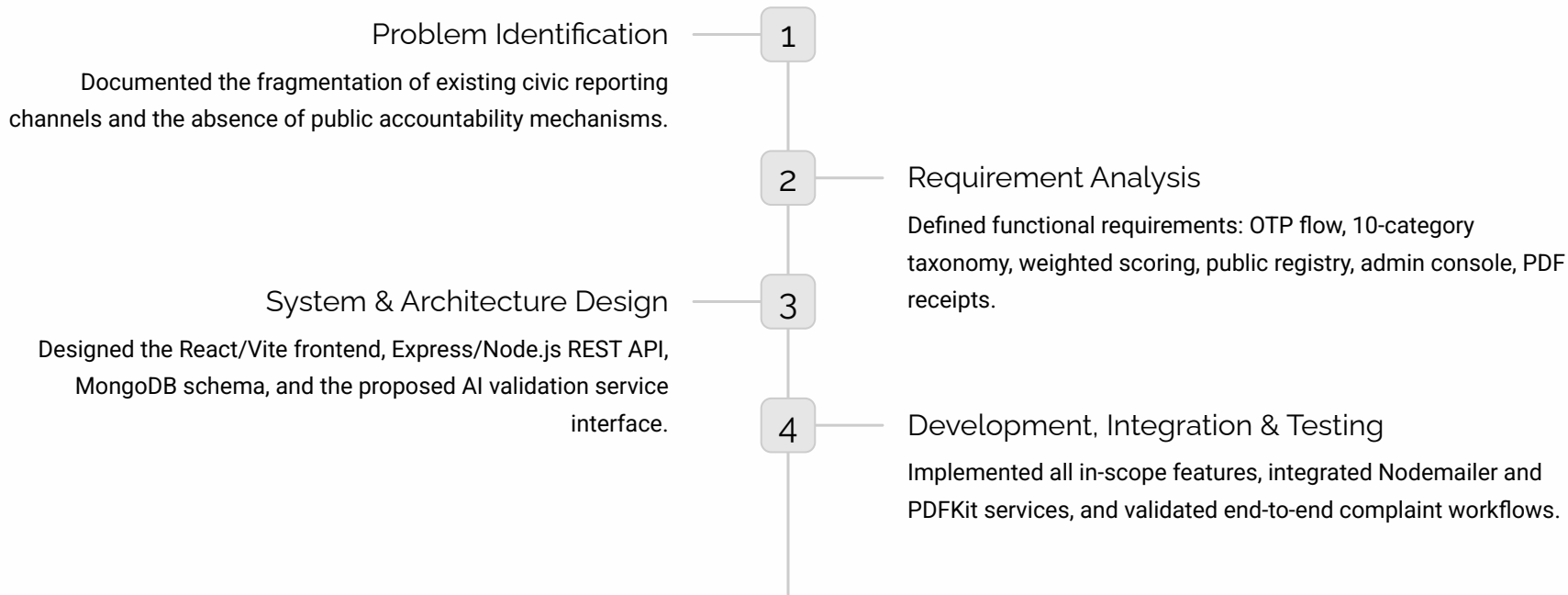
- Production AI validation service
- Automated complaint routing
- Analytics and reporting dashboard

Methodology

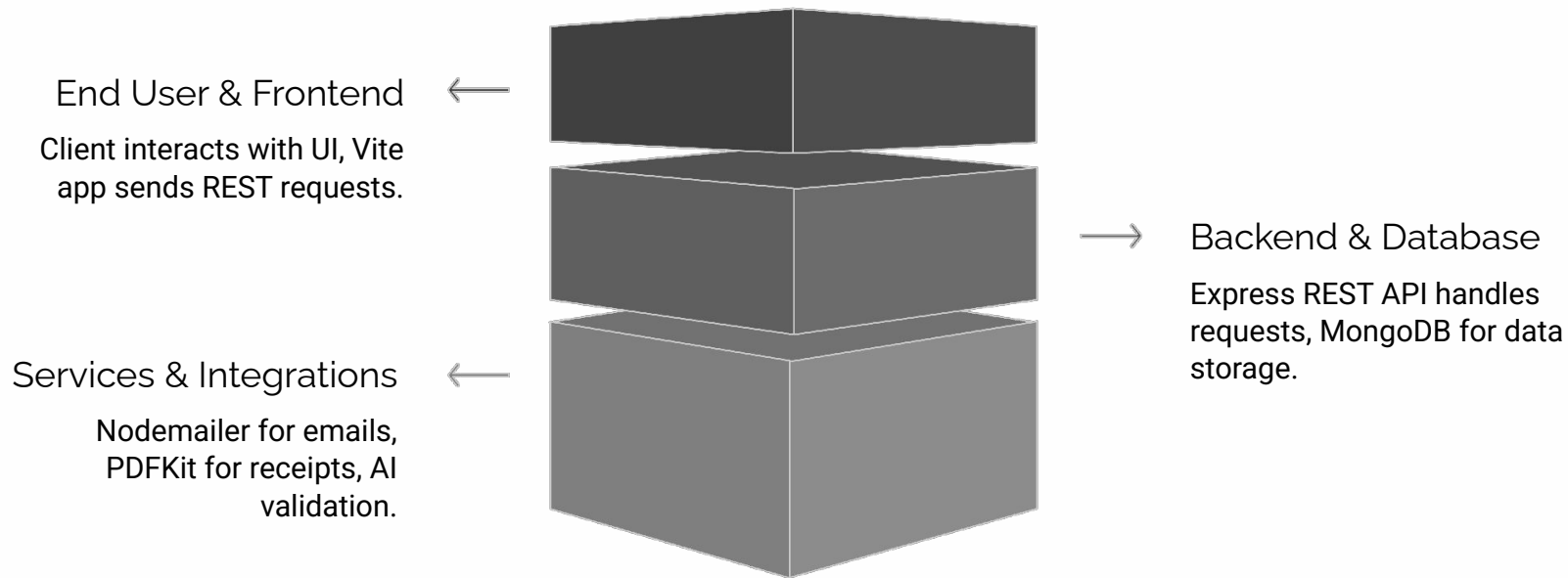
The project follows a structured, phase-gated development methodology – from problem identification through to deployment – ensuring each layer of the system is validated before the next is built.



Methodology



System Architecture



System Architecture

The system is composed of four implemented layers and one proposed extension. All inter-service communication is via REST API; the AI validation service is architecturally defined but not yet deployed.

Frontend — React (Vite)

Citizen complaint wizard, live priority score display, public complaint registry, and JWT-authenticated admin console.

Backend — Express.js / Node.js

REST API handling OTP issuance and verification, complaint CRUD, status workflow transitions, and service orchestration.

Database — MongoDB

Three primary collections: Users, Complaints (with full status history), and OTPs (TTL-indexed for automatic expiry).

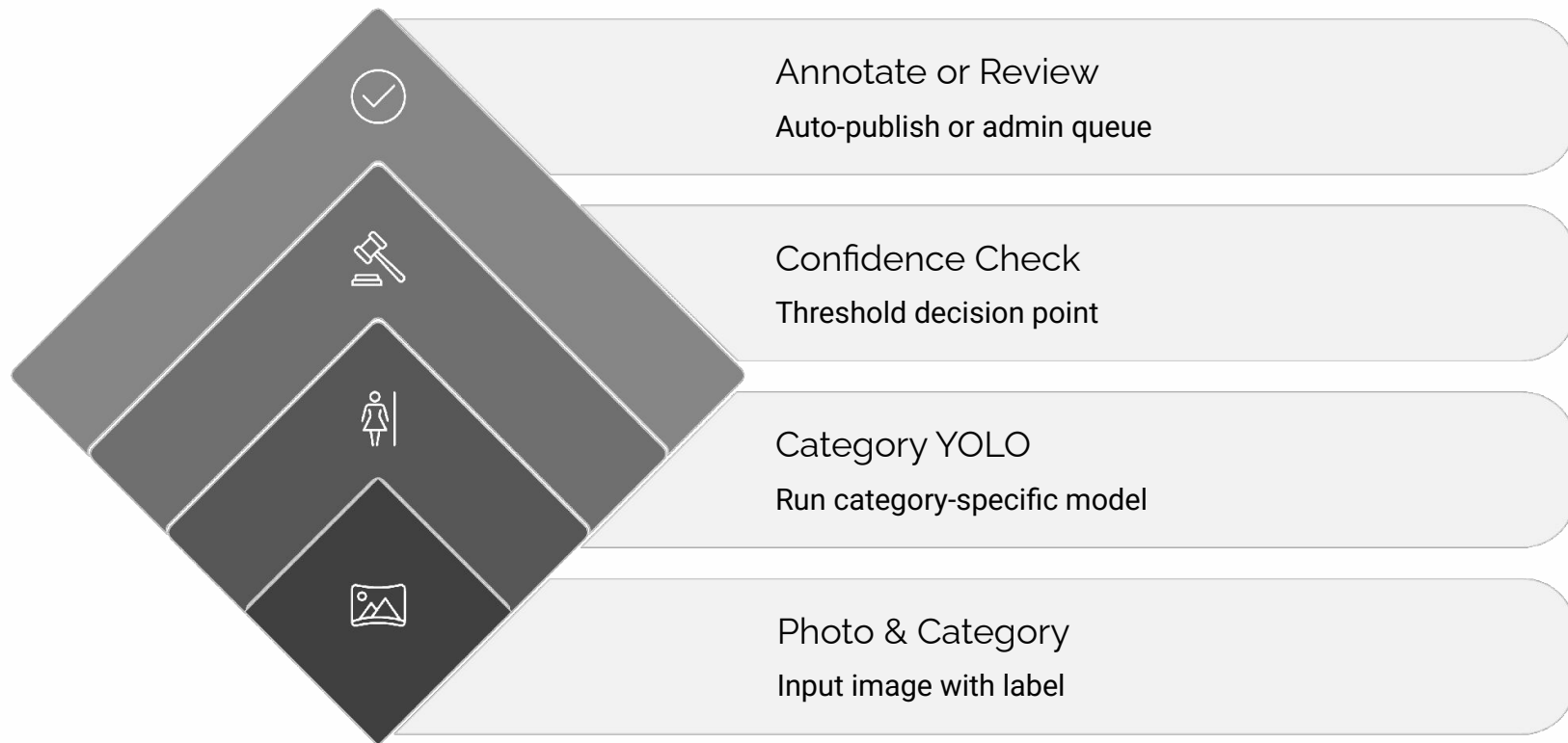
Services — Nodemailer + PDFKit

Nodemailer handles OTP delivery and status notification emails. PDFKit generates downloadable PDF receipts for each verified complaint.

AI Validation Service — *Proposed*

Architecturally defined as a sidecar service receiving photo + category from the backend and returning confidence score + annotated image. Not yet implemented in the current codebase.

AI Validation Layer



AI Validation Layer

The AI validation layer is designed to automate photo verification on a per-category basis, reducing manual admin review load while maintaining a human-in-the-loop fallback for low-confidence detections.

Model Design

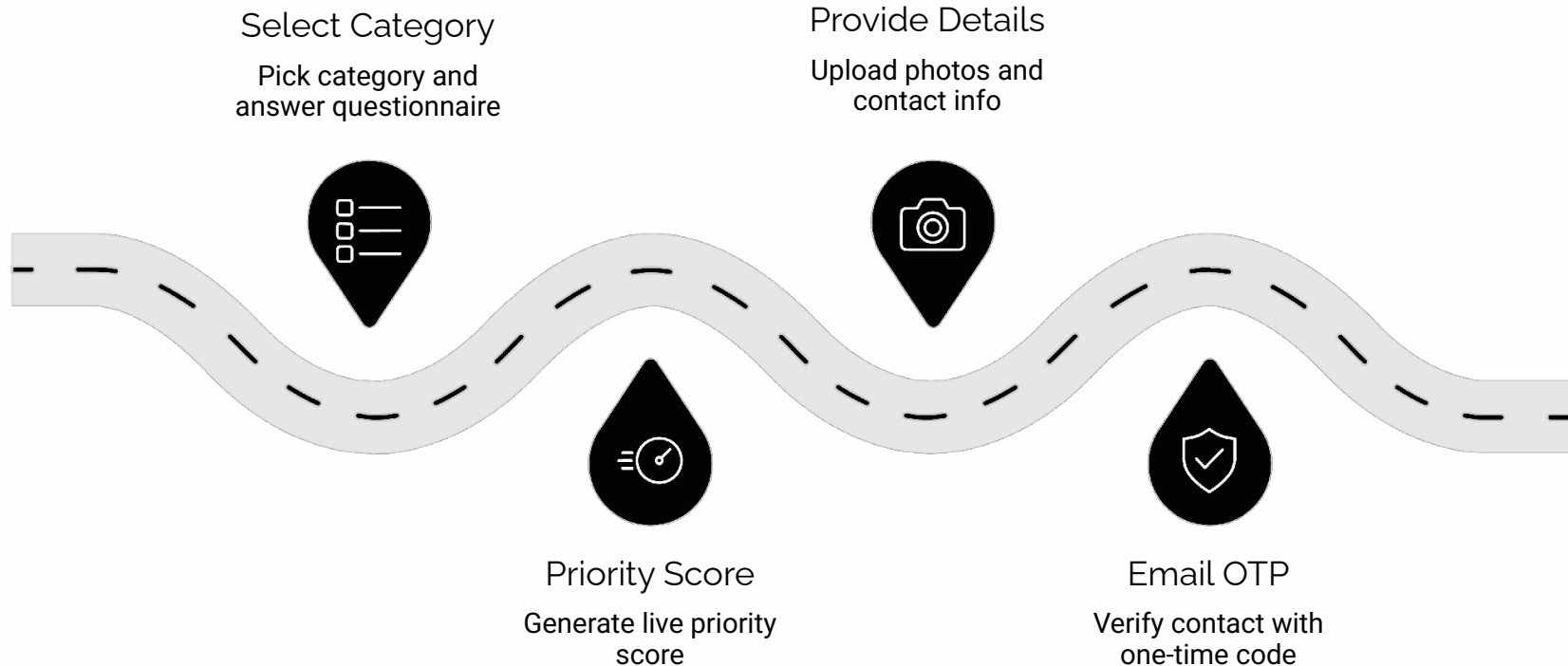
- One YOLO-based object-detection model per complaint category
- Example: `pothole_road_damage_model.pt`
- Each model trained on category-specific imagery
- Returns: confidence score + bounding-box annotated image

Routing Logic

- **High confidence:** annotated image auto-published with complaint
- **Low confidence:** complaint routed to admin review queue
- Admin can override, approve, or reject flagged submissions
- Maintains human accountability at every decision point

User Workflow

The citizen-facing workflow is designed to be completable in a single session with no prior registration. Each step produces a verifiable artifact — a priority score, a tracking ID, and a public registry entry.



User Workflow

1

Category + Questionnaire

Citizen selects one of 10 categories and completes the weighted questionnaire. Score is computed live.

2

Details + Photos

Citizen provides location details, description, and photo evidence. Contact email is collected for OTP delivery.

3

OTP Verification

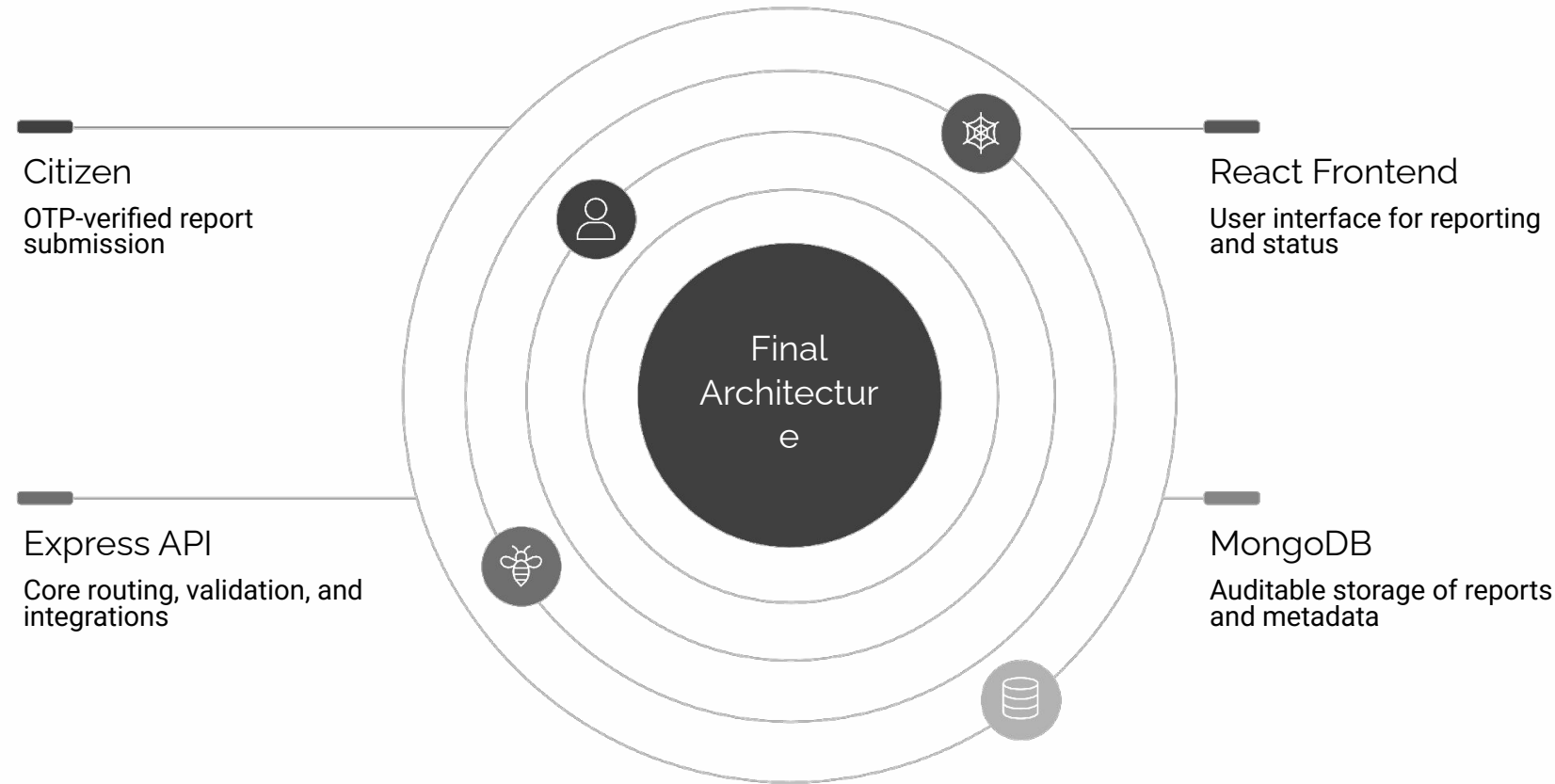
A one-time passcode is sent to the provided email. Submission is only finalized upon successful OTP entry — no account created.

4

Tracking ID + Public Registry

A unique tracking ID is issued. The complaint appears immediately in the public registry with full status history visible to all.

Conclusion & Final Architecture



Conclusion & Final Architecture

UrbanFix replaces fragmented, untracked civic reporting with one transparent, OTP-verified, auditable platform. The weighted questionnaire delivers consistent, explainable priority triage. The proposed AI layer adds automated visual validation as the system's next extension.

Problem Solved

Eliminates lost/duplicate complaints, inconsistent triage, and the absence of public accountability that characterize current reporting channels.

Implemented System

React/Vite frontend, Express/Node.js REST API, MongoDB with TTL OTPs, Nodemailer, PDFKit receipts, JWT-protected admin console, and public registry.

Proposed Extension

Per-category YOLO-based AI validation service — architecturally defined, with confidence-based routing to auto-publish or admin review queue.

Key Technical Distinctions

Implemented: OTP submission, 10-category taxonomy, weighted questionnaire scoring, public registry, admin status workflow, PDF receipts.

Proposed: YOLO-based per-category AI validation service with confidence-threshold routing.

Academic Contribution

Demonstrates that a zero-registration, transparency-first civic platform with severity-aware triage is achievable with a lean open-source stack — and defines a clear, implementable path to automated visual validation.